

David Rosocha

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EDUCATION

- University of Ottawa** Ottawa, Canada
 - BASc Electrical Engineering & BSc Computer Science (Double Major)* *Expected Dec. 2027*
 - **GPA: 3.99/4.00** (CGPA: 9.88/10.0)
 - Awards: Faculty of Engineering Co-op Student of the Year 2025, Director of Education Graduate Student Award

TECHNICAL SKILLS

- **Verification:** UVM, UVVM, SystemVerilog Assertions, constrained-random stimulus, functional coverage & closure, scoreboards, golden reference models, RTL simulation
- **Hardware:** SystemVerilog, Verilog, VHDL, RTL & FSM design, fixed-point arithmetic, AXI/AXI-Stream, I2C, UART, timing closure, FPGA implementation
- **Tools & Software:** Vivado, Questa, ModelSim, Quartus, Git, Linux, TCL; Python (NumPy, PyTorch), C/C++, MATLAB/Simulink

EXPERIENCE

- Ciena** Ottawa, Canada
 - FPGA Design & Verification Co-op* *Jan 2026 – Aug 2026*
 - Developed **UVVM verification environments** for the FPGA logic behind Hyper Rail, Ciena's flagship inter-data-center interconnect solution, verifying functional correctness, interface compliance, and timing behavior
 - Built constrained-random stimulus and protocol-level agents to drive **coverage closure** across core IP blocks, catching functional and timing bugs pre-silicon
 - Built a hardware tracer core in SystemVerilog capturing and timestamping up to **256 live signals at sub-10ns resolution**, with triggers on signals, vectors, and peripheral protocols (AXI, I2C), replacing commercial ChipScope tooling
 - Implemented AXI-Streaming and MPI interfaces for DMA transfer between the tracer core and an external CPU, which put real-time signal analysis on Linux workstations
- ShareLab, Carleton University** carleton.ca/share Ottawa, Canada
 - Undergraduate Researcher, Neuromorphic Hardware* *Sept 2026 – Present*
 - Designing an event-driven audio encoder in RTL using spiking neural networks, targeting FPGA implementation for a conference submission
 - Implemented a **multiplier-free leaky integrate-and-fire neuron** in SystemVerilog by constraining the discretized decay coefficient to a power of two, reducing the datapath to adders, a shift, and a comparator, inferring **zero DSP slices**
- Telesat** Ottawa, Canada
 - LEO Satellite Bus & Systems Engineering Co-op* *May 2025 – August 2025*
 - Designed a satellite coverage simulation tool in Python (NumPy/pandas), achieving a **300% runtime improvement** for 1400+ LEO satellites and replacing commercial tooling (e.g., STK)
 - Built an Optical Inter-Satellite Link framework simulating azimuth, Doppler shift, and alignment across **entire constellation orbit configurations**, informing crosslink feasibility
- uODD** uofpga.ca Ottawa, Canada
 - Founder* *Dec 2024 – Present*
 - Founded the first digital design club at uOttawa, growing to **160+ members in 8 months** by teaching Verilog and FPGA development
 - Coordinated FPGA workshops and authored open-source curriculum with **Ciena and Synopsys**

KEY PROJECTS

- FPGA Transformer Accelerator (TPU)** [GitHub](#)
 - SystemVerilog/UVM/Vivado/Questa/PyTorch* *2026*
 - Designed and verified a TPU-style transformer inference accelerator on a Basys 3 (Artix-7) FPGA, offloading the attention layers of a trained neural network to hardware over a UART host interface
 - Built a **UVM environment** from scratch (agent, sequencer, driver, monitor, scoreboard, constrained-random stimulus, and functional coverage), checking the datapath against a golden reference model to catch silent fixed-point errors
 - Implemented an **8×8 output-stationary systolic array** (64 MAC processing elements) with a double-buffered tiling controller scaling arbitrary matrix multiplies up to 64×64, sized to the board's 90 DSP slices
 - Designed a divider-free **hardware softmax** using a 2D-LUT approximation, validated to <0.9% mean error against a floating-point reference in Q8.8 fixed-point